databricks

Advance Lakeflow Jobs Features

LECTURE

Monitoring Jobs Performance

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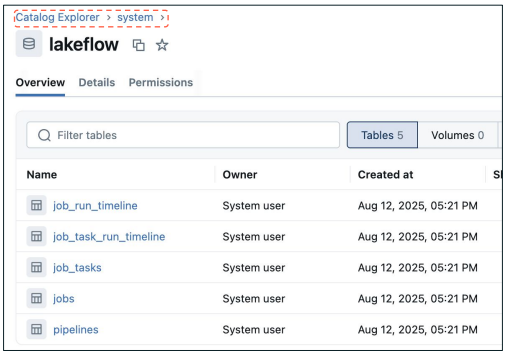
Failure handling isn't just about restarting tasks - it's about building resilient systems that can recover efficiently and maintain data consistency even when components fail.

Monitoring Jobs Performance

System Tables

- **system.lakeflow** is a built-in, read-only catalog that logs all job activity across workspaces in the region
- Timeline tables slice long runs hourly using **period_start_time** and **period_end_time**, enabling reliable duration, concurrency, and SLA analytics.
- Key Tables:
  - **jobs** (Job basic info)
  - **job_tasks** (Task basic definitions)
  - **job_run_timeline** (Each job run over time)
  - **job_task_run_timeline** (Each task run over time)
  - **Pipelines** (Pipelines basic info)



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The `system.lakeflow` catalog provides enterprise-grade monitoring capabilities:

Comprehensive Logging: All job activity across all workspaces in the region is automatically logged, providing complete visibility into workflow execution patterns and performance trends.

Timeline Analysis: Timeline tables use `period_start_time` and `period_end_time` to slice long-running jobs into hourly segments, enabling accurate duration analysis, concurrency tracking, and SLA measurement even for complex, long-running workflows.

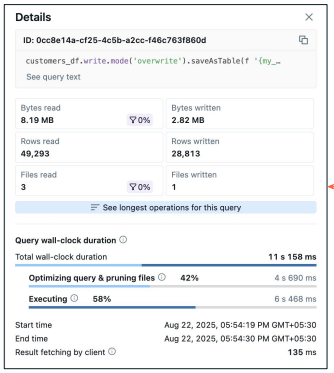
Key Table Functions:

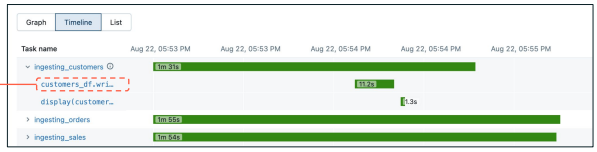
- jobs: Basic job metadata and configuration information
- job_tasks: Task definitions and configuration details
- job_run_timeline: Complete execution history for every job run
- job_task_run_timeline: Detailed execution history for individual tasks
- pipelines: Information about Delta Live Tables pipelines

Analytics Capabilities: This data enables sophisticated analytics including cost analysis, performance trending, SLA compliance tracking, and resource utilization optimization.

Monitoring Jobs Performance

Spark UI





- **Timeline** in job run highlights task start/end, duration, and overlap to spot bottlenecks fast.
- Click a task to see status, timestamps, duration, cluster/runtime, logs, and quick I/O.
- Click on **query/code** details for full text, run ID, wall-clock split (optimizing/pruning vs executing), files read/written details, files & partitions and spill details

Act on insights:
High planning time → Improve pruning/partitioning
High execution time → Optimize joins/aggregations (broadcast, skew fixes)

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The Spark UI provides detailed performance insights for optimization:

Timeline Analysis: The execution timeline immediately highlights task duration, overlap, and bottlenecks, enabling quick identification of performance issues.

Task-Level Details: Clicking individual tasks reveals comprehensive information including execution timestamps, resource utilization, cluster configuration, logs, and I/O statistics.

Query Performance Details: Detailed query analysis shows execution plans, optimization decisions, file access patterns, partition information, and data spill details - essential for performance tuning.

Actionable Insights:

- High Planning Time: Indicates need for better partitioning strategies or metadata optimization
- High Execution Time: Suggests opportunities for join optimization, broadcast strategies, or skew handling
- Resource Bottlenecks: Identifies memory, CPU, or I/O constraints that require cluster configuration adjustments

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